MILITARY SPECIFICATION SHEET

ELECTRON TUBE, TRANSMITTING

TYPE 837

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentode, transmitting

Outline	---	16-2 (EIA)
Base	---	A7-13, low-loss phenolic
Envelope	---	ST-16
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	Cap
Element	---	h	sd	g2	g1	g3	k	h	a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ec3	Ib	Ic1	Pc2	Pc3	Pp	Pi	Ehk	Modulation	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	mAdc	mAdc	W	W	W	W	v		ft
Maximum:														
B Telep:	12.6 ±10%	500	---	200	200	40	---	5	5	12	16	100	---	10,000
C Telep:	12.6 ±10%	500	-200	200	---	40	8	8	---	12	16	100	g3	10,000
C Telep:	12.6 ±10%	400	-200	200	200	50	8	5	5	8	20	100	anode	10,000
C Teleg:	12.6 ±10%	500	-200	200	200	80	8	8	5	12	32	100	---	10,000
TEST CONDITIONS:	12.6	500	-16	200	0	---	---	---	---	---	---	---	---	---

GENERAL:

First article inspection - Required (See note 1)

Ⓒ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1266	Total grid current	2,3	Ec1/Ib = 36 mAdc; t = 300	0.65	II	Ic1	0	-5.0	µAdc
1256	Electrode current (1) (anode)	4		0.65	II	Ib	15	33	mAdc
1256	Electrode current (2) (anode)	-	Ec3 = -145 Vdc	0.65	II	Ib	2	12	mAdc
1231	Emission	-	Eb = Ec1 = Ec2 = Ec3 = 30 Vdc	0.65	II	Is	130	---	mAdc
1236	Power oscillation (1)	-	Ec3 = 60 Vdc; Rg1 = 10,000 ohms; Ic1 = 5 to 7 mAdc; Ib = 60 mAdc; Ic2 = 17 to 35 mAdc; F = 15 MHz	0.65	II	Po Ic3	15 4	--- ---	W mAdc
1201	Short and discontinuity detection	-		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1301	Filament current	-		---	---	If	0.63	0.77	A
1336	Heater-cathode leakage	-		---	---	Ihk	---	100	uAdc
1331	Direct-interelectrode capacitance	-	Shield No. 312 Without shield Without shield	6.5	Code E	Cga Cin Cout	--- 13 7.5	0.20 19 12.5	pF pF pF
1031	Low-frequency vibration	-	Eb = 250 Vdc, Ec2 = 100 Vdc; Ec1 = -10 Vdc; Rp = 2,000 ohms	---	---	Ep	---	500	mVac
1101	Secureness of base and cap	-		---	---	---	---	---	---
1111	Base pin solder depth	5		---	---	---	---	---	---
1211	Insulation of electrodes	-		4.0	S3	R	10	---	Meg
1036	Bump	-	Angle = 25°	6.5	S4	---	---	---	---
1105	Permanence of marking	-		---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test	-	Group B; Ec1 = -20 Vdc; Ec2 = 250 Vdc; Ehk = 100 V	---	---	---	---	---	---
---	Life-test end point (500 hours):								
1231	Emission	-		---	---	Is	100	---	mAdc
(B) 1236	Power oscillation (2)	6	Ec3 = 60 Vdc; Rg1 = 10,000 ohms; Ec1 = 5 to 7 mAdc; Ib = 60 mAdc; Ic2 = 17 to 35 mAdc; F = 20 MHz	---	---	Po	15	---	W

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NOTES:

1. All tests applicable herein, including quality conformance inspection parts 1, 2, and 3, shall be performed during first article inspection. The number of units to be tested shall be as specified in the contract or order.

2. Approved alternate test: Method 1266 (total grid current) Ecl = 8 Vdc, Icl = 5 µAdc maximum.

3. This test to be performed at the conclusion of the holding period.

4. Approved alternate test: Method 1256 (electrode current (annode)) ECl = -8 Vdc, Ib = 36 mAdc minimum to 70 mAdc maximum.

5. This note applies for flexible leads as well as for rigid leads.

(B) 6. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 80
DSA - ES

User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3103)

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A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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☐ YES ☐ NO *(If "Yes", in what way?)*

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